Research Hypothesis

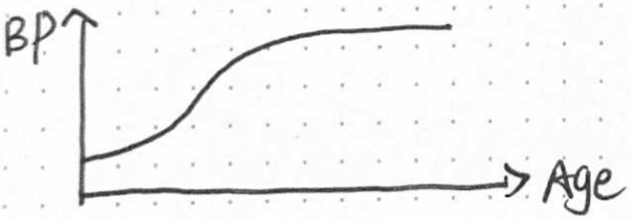
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1. Model form : polynomial vs. sigmoid / Gaussian.

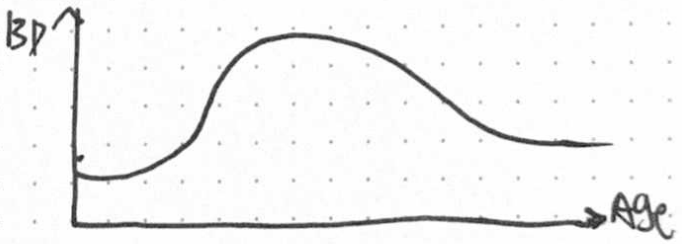
I expect a sigmoid to fit SBP better because SBP may rise with age and then level off.

A Gaussian should fit DBP better because DBP may peak around middle age. These shapes are more plausible than unrestricted quadratic trends. they may have lower MSE, although BIC could favor simpler trends.

SBP : sigmoid trend



DBP : Gaussian trend.



$SBP = S_0 + \frac{\Delta S}{1+e^{-\frac{age-age_0}{s}}}$ $DBP = D_0 + \Delta D e^{-\frac{(age-age_c)^2}{2\sigma^2}}$.

2. Effect of increasing class imbalance

As the male fraction rises from 50% to 97%, I expect more predictions of the majority class: male. recall will rise while female recall will fall. Accuracy may increase on the imbalanced test sets, but the balanced test set should reveal the loss of female recognition.

3. predictions with 90% male training data.

With 90% male / 10% female training data, female samples in the overlapping SBP / DBP region will often be predicted as male. I expect class reweighting or realistic sythetic female augmentation to reduce the recall gap, possibly at some cost to overall accuracy.